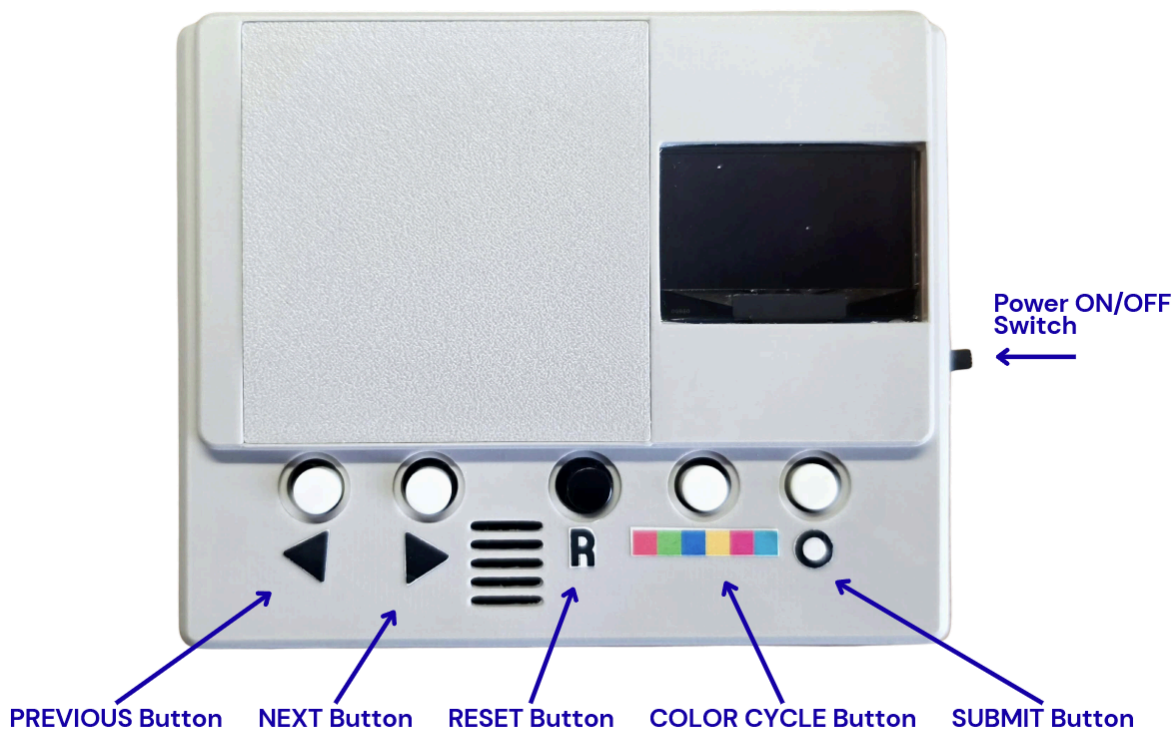


User Manual

RGBreak the Code: Handheld Logic Puzzle Device



A. Operation's Manual

1) System Requirements

The *RGBreak the Code* console is a standalone embedded device. It operates without network connectivity or an external processor. System requirements are listed in Table VII.

TABLE I.
HARDWARE AND SYSTEM REQUIREMENTS

Component	Specification
Core Hardware	ESP32-S3 Super Mini microcontroller, multiplexed LED matrix, OLED display, tactile push buttons.
Power Supply	Rechargeable 803860 LiPo battery (3.7V) housed in a modular cartridge, supported by a standard 5V USB-C charging interface (<i>Note: High-wattage PD chargers are not supported</i>).
Maintenance	PC environment running VS Code with PlatformIO, connected via USB-C data cable (Required strictly for firmware flashing).

2) Installation Procedures

- Verify presence of enclosure, ESP32-S3, WS2812 LED matrix, OLED display, pushbuttons, and 803860 LiPo battery.
- Install ESP32-S3 and WS2812 LED matrix in the enclosure.

- Place the diffuser plate over the LED matrix. Align OLED display with front panel.
- Connect I2C for OLED, GPIO for buttons and WS2812 data lines as per schematic.
- Insert 803860 LiPo battery in cartridge and connect to power.
- Secure the cartridge lid on the rear of the device and push until the slide-to-lock mechanism engages.
- Connect the ESP32-S3 to the computer via USB-C and upload firmware.
- Switch on power and confirm ESP32-S3 initializes and OLED displays main menu.

3) User's Manual

a) Menu Configuration

The OLED displays the graphical user interface (GUI) at startup. Navigation uses NEXT/PREV buttons. SUBMIT confirms selections. Game parameters are listed in Table II.

TABLE II.
GUI CONFIGURATION PARAMETERS

Parameter	Options	Description
Mode	Random, Repeat, Unique	Rules for secret code generation,

		including duplicate color settings.
Chances	3, 4, 5, 10	Number of guesses allowed per game.
Timer	OFF, 60s, 90s	Time limit for completing the puzzle.
Help	Selectable Interface	Digital instructions for gameplay.

b) Gameplay Loop

- Select *Start Game* from the main menu.
- Use NEXT/PREV buttons to move the cursor across the four square-shaped LED slots in the active row. Each row is a separate evaluation attempt.
- Press the COLOR CYCLE button to iterate through the six distinct color states for the localized slot..
- Press SUBMIT button to lock the sequence for that row and trigger the evaluation algorithm.

c) Evaluating Feedback

Upon submission, the WS2812 matrix displays feedback for the attempt on the same row. White indicates an exact match (correct color and position). Orange indicates a partial match (correct color, wrong position). If unsolved, the game advances to the next row for another attempt, up to the guess limit.

d) Game Termination

To abort a session, press the RESET button. The system will pause the game and display a YES/NO confirmation on the OLED. Selecting YES ends the evaluation loop, clears memory, and returns to the main menu.

4) Troubleshooting Guides and Procedures

TABLE III.
HARDWARE AND SYSTEM TROUBLESHOOTING GUIDE

Edge Case	Cause	Procedure
Unresponsive / Blank OLED	System boot failure, loose terminal, or total power loss.	Verify the LiPo battery is seated and connected. If so, recharge the device.
Charging Failure (High-Wattage)	USB-PD handshake failure.	Use a standard 5V ($\leq 25W$) adapter or USB-A to USB-C cable. Avoid high-wattage or smart chargers.

Dim / Uneven LED Diffusion	Localized voltage drop (LED matrix sensitivity).	Check the OLED battery indicator. Recharge to restore voltage and brightness.
Unresponsive buttons	Hardware debouncing fault or ESP32-S3 logic loop hang.	To reset, toggle the power switch OFF, then ON. This reboots the system and clears memory.

5) Error Definitions

TABLE IV.
PROGRAMMED ERROR DEFINITIONS AND RESPONSES

Error State	Trigger Condition	System Responses
Incomplete Row Error	The user presses SUBMIT with unassigned slots in the sequence.	OLED displays "FILL ALL COLORS" warning and the buzzer sounds error cue. Submission is rejected, cursor row remains, and guess is not deducted.
Low Power Warning	Battery voltage drops below minimum threshold.	The battery icon shows empty. Recharge to avoid shutdown and data loss.
Game Termination Prompt	The user presses RESET during an active loop.	System pauses and shows YES/NO confirmation on OLED before resetting to prevent data loss.

B. Pictures of Prototype



Fig. 1. Assembled RGBreak the Code handheld console with custom 3D-printed enclosure.

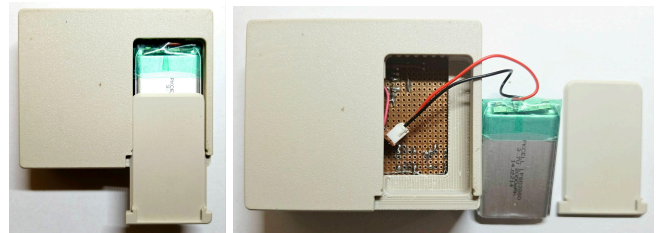


Fig. 2. Battery cartridge with slide access panel and LiPo battery connection.



Fig. 3. OLED interface showing game menu (left) and game instruction (right). The GUI supports selection of logic modes, guess limits, time constraints, and displays battery level.



Fig. 4. OLED display showing error alert for incomplete row submission.



Fig. 5. LED matrix feedback is shaped as circles showing exact match indicators (white: correct color and position) and partial match indicators (orange: correct color, incorrect position) after evaluation.



Fig. 6. End-game state showing generated four-color secret code.

C. Data Sheets

TABLE VI.
TABLE OF COMPONENTS AND SPECIFICATIONS

Component	Specification
ESP32-S3 Super Mini	512 KB SRAM, 384 KB ROM built-in, with 4 MB Flash
TP4056 Charger Module	Preset 4.2 V Charge Voltage, 2.9 V Trickle Charge Threshold

MT3608 Boost Converter Module	2-24 V Input Voltage, Up to 28 V Output Voltage
803860 LiPo Battery	2200 mAh, 3.7 V
WS2812 8-Bit 5050 LED Strip Board Modules	4-7 V, 18 mA current drive
SH1106 OLED Display	132 X 64 bits SRAM
Momentary Tactile Push Buttons	6 x 6 x 7 mm, 4-pin
Metal Film Resistors	100 k Ω , 1/4 Watts, 1% tolerance
OBO-152 Piezoelectric Passive Buzzer	80 dB minimum at 4 kHz 9 V, 3 mA maximum, 16.5 mm diameter
SK-22D05 2P2T Switch	0.3 A at 30 V, 30 m Ω maximum
JST Connectors	2.54 mm pitch terminal, 2-pin
Copper Plate Bakelite Perfboard	90 x 70 mm, 1.2 mm thickness
Custom Enclosure	3D-printed PLA matte

TABLE VII.
3D-PRINTED ENCLOSURE DIMENSIONS

Component	Dimensions (L x W x H)
External Housing	105.0 x 85.2 x 31.2 mm
Internal Housing	92.1 x 72.1 x 15.1 mm
Internal Battery Cartridge	39.3 x 62.5 x 8.5 mm